

- 4 (a) The body temperature of a human is maintained at its set point of approximately 37°C. If it rises above this temperature, physiological responses begin to return the temperature to its set point. Two of these responses are vasodilation and sweating.

Explain how vasodilation and sweating help to return the body temperature to its set point.

vasodilation

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sweating

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- (b) Diabetes mellitus is a disease where the pancreas is not able to secrete sufficient insulin.

The symptoms of diabetes mellitus include a tendency to drink a lot of water and a loss of body mass.

Suggest why these symptoms occur.

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(c) A person with diabetes mellitus can use a biosensor to measure the concentration of glucose in their blood.

(i) Outline how a glucose biosensor works.

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(ii) Suggest **one** advantage of using a biosensor rather than a dip stick.

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[Total: 12]